ZHANG Bin, Ph.D. Candidate

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Education

Sep. 2021 – Now	**Ph.D. Student, The Hong Kong Polytechnic University** in *Mechanical Engineering*.
Mar. 2024 – Jul. 2024	**Visiting Ph.D. Student, KTH Royal Institute of Technology**.
Sep. 2017 – Jul. 2020	**M.Sc., China Academy of Space Technology** in *Control Science and Engineering*.
Sep. 2013 – Jul. 2017	**B.Eng., Beihang University** in *Detection, Guidance, and Control Technology*.

Employment History

May 2021 – Aug. 2021	**Research Assistant** at ROMI Lab, The Hong Kong Polytechnic University.

Research Interests

- Multi-agent formation control.
- Multi-robot manipulation.
- Control theory and applications.

Research Publications

* Equal contribution

Journal Articles

1. N. Han, G. Gong, **B. Zhang**, *et al.*, "Prescribed performance control of deformable object manipulation in spatial latent space," *IEEE/ASME Transactions on Mechatronics*, 2025, (Under Review).
2. X. Shao*, **B. Zhang***, H. Zhi, *et al.*, "Inptc: Integrated planning and tube-following control for prescribed-time collision-free navigation of wheeled mobile robots," *IEEE/ASME Transactions on Mechatronics*, 2024, (Early Access). DOI: 10.1109/TMECH.2024.3486727. URL: https://vimeo.com/895801720.
3. Y. Tian, **B. Zhang**, X. Shao, and D. Navarro-Alarcon, "Encircling general 2-d boundaries by mobile robots with collision avoidance: A vector field guided approach," *IEEE/ASME Transactions on Mechatronics*, 2024, (Under Review). DOI: arXiv:2501.02242.
4. **B. Zhang**, H. Zhi, J. G. Romero, and D. Navarro-Alarcon, "Leader-follower formation control of perturbed nonholonomic agents along parametric curves with directed communication," *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8603–8610, Aug. 2024. DOI: 10.1109/LRA.2024.3445657. URL: https://vimeo.com/872927697.
5. H. Zhi, **B. Zhang**, J. Qi, *et al.*, "Non-prehensile object transport by nonholonomic robots connected by linear deformable elements," *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8651–8658, Aug. 2024. DOI: 10.1109/LRA.2024.3440096.
6. H. Y. Lee, P. Zhou, **B. Zhang**, *et al.*, "A distributed dynamic framework to allocate collaborative tasks based on capability matching in heterogeneous multi-robot systems," *IEEE Transactions on Cognitive and Developmental Systems*, vol. 16, no. 1, pp. 251–265, Feb. 2023. DOI: 10.1109/TCDS.2023.3264034.

7. W. Ma, **B. Zhang**, L. Han, S. Huo, H. Wang, and D. Navarro-Alarcon, "Action planning for packing long linear elastic objects into compact boxes with bimanual robotic manipulation," *IEEE/ASME Transactions on Mechatronics*, vol. 28, no. 3, pp. 1718–1729, Jun. 2023. DOI: 10.1109/TMECH.2022.3223990.

8. M. Muddassir, G. Limbert, **B. Zhang**, A. Duan, J.-J. Tan, and D. Navarro-Alarcon, "Model predictive thermal dose control of a robotic laser system to automate skin photorejuvenation," *IEEE/ASME Transactions on Mechatronics*, vol. 28, no. 2, pp. 737–747, Apr. 2023. DOI: 10.1109/TMECH.2022.3218806.

9. **B. Zhang**, H. Zhi, J. G. Romero, *et al.*, "Fourier-based multi-agent formation control to track evolving closed boundaries," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 70, no. 11, pp. 4549–4559, Nov. 2023. DOI: 10.1109/TCSI.2023.3300772. URL: https://vimeo.com/809466573.

10. S. Huo, **B. Zhang**, M. Muddassir, D. T. W. Chik, and D. Navarro-Alarcon, "A sensor-based robotic line scan system with adaptive roi for inspection of defects over convex free-form specular surfaces," *IEEE Sensors Journal*, vol. 22, no. 3, pp. 2782–2792, Feb. 2022. DOI: 10.1109/JSEN.2021.3132428.

11. J. Zhou, J. Hu, Y. Bai, and **B. Zhang**, "Optimal impulsive time-fixed transfers around the libration points of the restricted three-body problem," *Astrophysics and Space Science*, vol. 365, pp. 1–18, May 2020. DOI: 10.1007/s10509-020-03793-7.

12. J. Zhou, J. Hu, and **B. Zhang**, "Analytical solutions for relative motion in the cicular restricted three-body problem," *Journal of Astronautics*, vol. 41, no. 2, Feb. 2020, (in Chinese). DOI: 10.3873/j.issn.1000-1328.2020.02.004.

13. J. Zhou, J. Hu, and **B. Zhang**, "Approximate analytical solutions of motion near the collinear libration-points in restricted three-body problem," *Journal of Deep Space Exploration*, vol. 7, no. 1, Feb. 2020, (in Chinese). DOI: 10.15982/j.issn.2095-7777.2020.20190408001.

14. **B. Zhang** and J. Zhou, "Characteristic model-based station-keeping control for halo orbit," *Acta Aeronautica et Astronautica Sinica*, vol. 40, no. 11, Nov. 2019, (in Chinese). DOI: 10.7527/S1000-6893.2019.23306.

Conference Proceedings

1. A. Duan, **B. Zhang**, L. Hu, L. Feng, and D. Navarro-Alarcon, "Balancing control of wheeled-legged hexapedal robots for traversing rough terrains," in *2022 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Dec. 2022, pp. 1905–1910. DOI: 10.1109/ROBIO55434.2022.10011895.

2. **B. Zhang**, C. Li, and J. Zhou, "Station-keeping of halo orbits using convex optimization-based receding horizon control," in *2021 China Automation Congress (CAC)*, Oct. 2021, pp. 7160–7165. DOI: 10.1109/CAC53003.2021.9728345.

3. **B. Zhang** and J. Zhou, "Characteristic model-based station-keeping control for halo orbits about sun-earth l1 point," in *2019 Chinese Automation Congress (CAC)*, Nov. 2019, pp. 418–423. DOI: 10.1109/CAC48633.2019.8997351.

Skills

Languages	English, Chinese (*mother tongue*).
Coding	Python, C/C++, Matlab, HTML, LaTeX.
Others	ROS, Linux, Webots, Arduino, SolidWorks.

Miscellaneous Experience

Awards and Achievements

2017 **Outstanding Graduate of Beihang University.**

2015 **National Inspirational Scholarship.**

2014 **National Inspirational Scholarship.**

Academic Services

2021 **Reviewer**. ICRA 2022.

2023 **Reviewer**. ICRA 2024.

Reviewer. IEEE Control Systems Letters.

2024 **Reviewer**. Journal of Field Robotics.

Reviewer. IEEE Robotics and Automation Magazine.

Reviewer. IEEE Transactions on Control of Network Systems.

Reviewer. CASE 2024.

Reviewer. ICRA 2025.

Reviewer. Aerospace Science and Technology.

2025 **Reviewer**. IEEE Transactions on Robotics.

Teaching Experience

2021 **Teaching Assistant**, Perceptual Robotics.

2022 **Teaching Assistant**, Linear Systems and Control.

Teaching Assistant, Advanced Control Technology.

2023 **Teaching Assistant**, Linear Systems and Control.